The Universal Power-of-Two Basis

Transcendental Constants as Single Integers Over 2^{335} at Sub-Planck Precision

Registry: [HOWL-MATH-4-2026]

Series Path: [HOWL-MATH-2-2026] → [HOWL-MATH-3-2026] → [HOWL-MATH-4-2026]

DOI: 10.5281/zenodo.zzz

Date: March 31 2026

Domain: Computational Mathematics / Exact Arithmetic / Number Theory

Status: Complete

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I. ABSTRACT

[HOWL-MATH-2-2026] established that 17 transcendental constants can be represented as exact integer pairs (p, q) at 100-digit precision using convergent rational series in exact arithmetic. Each constant has a different denominator q, determined by the series used to compute it. Operations between constants — addition, subtraction, comparison — require computing least common denominators between large integers, which is computationally expensive and produces intermediate values with inflated digit counts.

This paper shows that all 22 constants in the extended MATH-2/MATH-3 basis can be represented as single integers over a shared denominator 2^{335}, verified at 100 digits against mpmath references. The choice of a power-of-two denominator is motivated by the continued fraction structure of Euler's number: 87/32 is the 5th convergent of e, the provably best rational approximation with denominator ≤ 32, and $32 = 2^5$. Extending to 2^{335} provides 100-digit precision for every constant in the basis.

Under this representation, addition and subtraction of any two transcendental constants reduces to addition or subtraction of their integer numerators. No least common denominator computation is required. The shared denominator 2^{335} is stored once. The total storage for 22 constants is 2,238 digits plus the exponent 335 — compared to approximately 20,000 digits for the equivalent MATH-2 pairs. The compression ratio ranges from 2.3 × for e to 1,280 × for e^ π .

The representation is a change of encoding, not new mathematics. The constants are the same. The precision is the same. The sub-Planck threshold argument from MATH-2 applies identically. The contribution is the observation that a single power-of-two denominator serves the entire basis, and that this encoding optimizes the arithmetic operations

most commonly performed in physics calculations: linear combinations of transcendentals with rational coefficients.

II. THE CONTINUED FRACTION ORIGIN

2.1 The Observation

The number 87/32 is close to Euler's number: 87/32 = 2.71875, e = 2.71828…, a difference of 0.017%. This proximity is not accidental. It is a consequence of the continued fraction expansion of e.

The continued fraction of e is:

e = [2; 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, …]

with the well-known pattern: the CF coefficients are [2; 1, 2k, 1] repeating for k = 1, 2, 3, …. The convergents of this CF are the provably best rational approximations to e — no fraction with a smaller or equal denominator is closer.

2.2 The Convergent Table

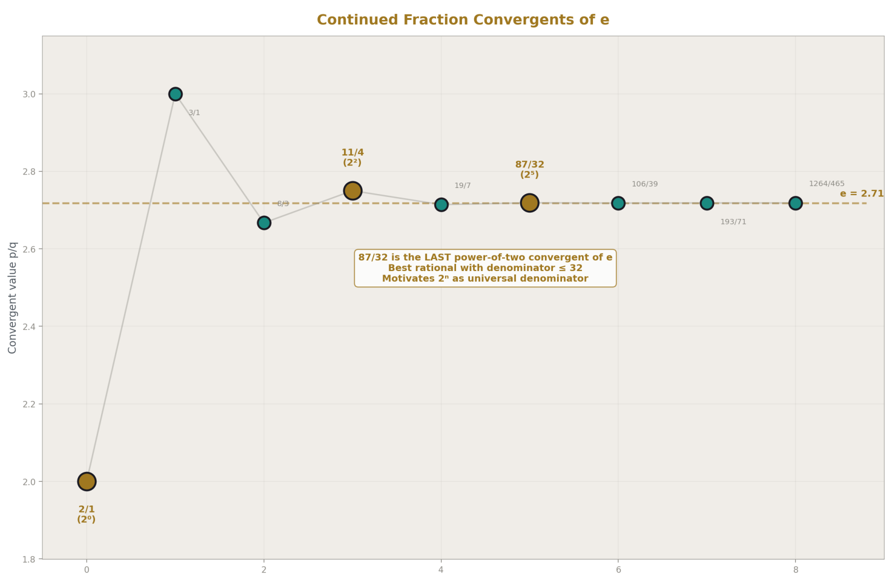

Figure 1: Fig. 1: CF convergents of e — three have power-of-two denominators. 87/32 is the last and best.

Index	CF coeff	p	q	Decimal	Error (ppm)
0	2	2	1	2.000	264,241
1	1	3	1	3.000	103,638
2	2	8	3	2.667	18,988
3	1	11	4	2.750	11,668
4	1	19	7	2.714	1,470
5	4	87	32	2.71875	172
6	1	106	39	2.71795	123
7	1	193	71	2.71831	10.3
8	6	1264	465	2.71828	0.83

87/32 is convergent [5]. It is the best rational approximation to e with denominator ≤ 32. The denominator $32 = 2^5$ is a power of two.

2.3 Power-of-Two Convergents

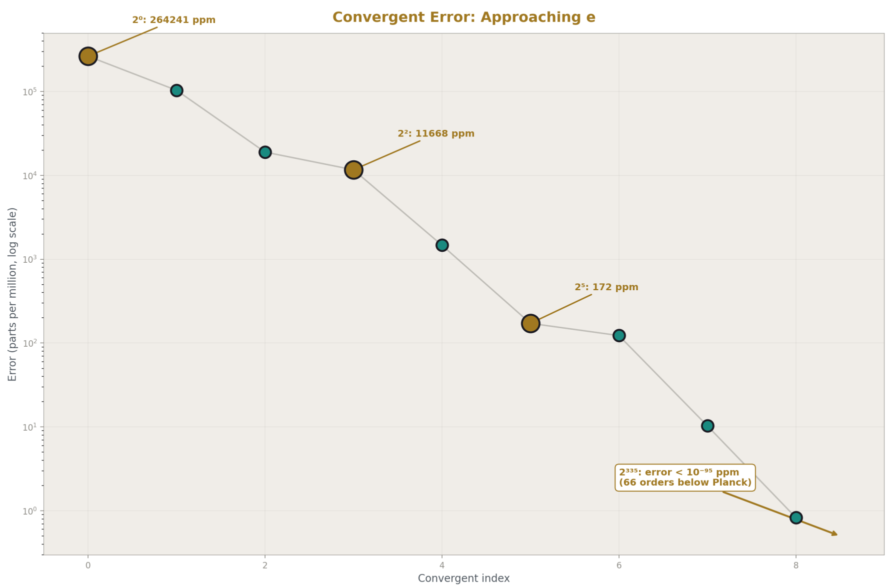

Figure 2: Fig. 2: Error drops from 264,000 ppm to <1 ppm across 9 convergents. 87/32 at 172 ppm is the last power-of-two.

Three convergents of e have power-of-two denominators:

Convergent	Fraction	Denominator	ppm
[0]	2/1	2^0	264,241
[3]	11/4	2^2	11,668
[5]	87/32	2^5	172

The exponents 0, 2, 5 do not continue — no further CF convergent of e has a power-of-two denominator in the first 25 convergents. The 87/32 convergent is the last and best power-of-two convergent of e at human-readable scale.

2.4 The Extension

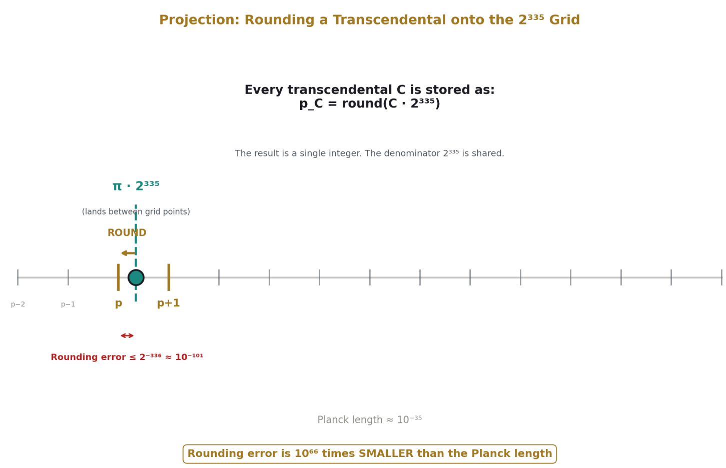

Figure 3: Fig. 4: Each transcendental is rounded to the nearest integer multiple of 2^{-335}. The rounding error is 10^{66} times smaller than the Planck length.

The CF convergent observation motivates the question: at what power of two does the nearest integer to $e \cdot 2^n$ provide 100-digit agreement with e?

The rounding error when projecting a real number x onto the nearest $p/2^n$ is bounded by $1/(2 \cdot 2^n) = 2^{-(n+1)}$. For 100-digit agreement, we need $2^{-(n+1)} < 10^{-100}$, giving $n > 100 \cdot \log_2(10) - 1 \approx 331.2$. Constants smaller than 1 (such as $\ln(2) \approx 0.693$) require an additional bit because their leading digit occupies less of the available range.

Empirically, n = 335 provides 100-digit agreement for all 22 constants in the extended MATH-2/MATH-3 basis. At n = 334, one constant (Catalan's G) fails at position 101. At n = 333, two fail. At n = 335, all 22 match.

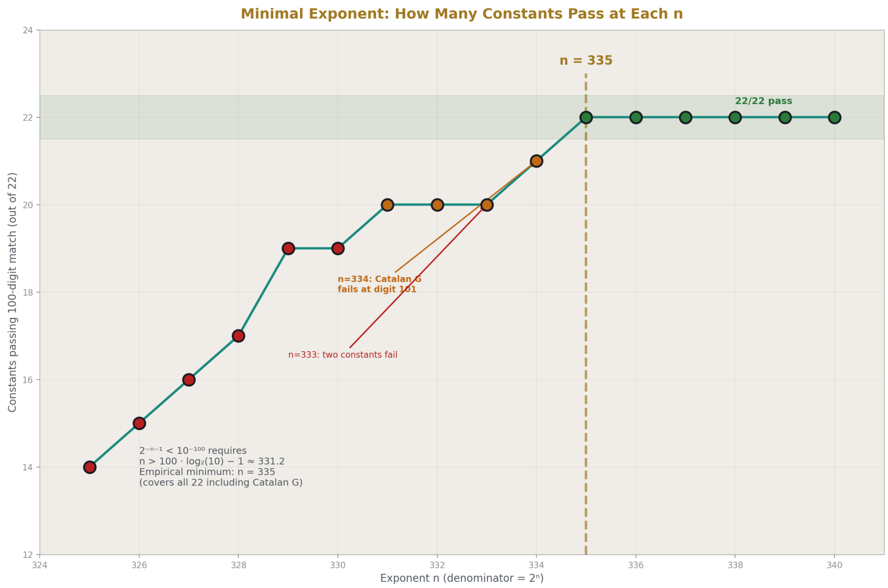

Figure 4: Fig. 7: At n=334, Catalan's G fails. At n=335, all 22 constants pass 100-digit verification. 335 is the minimal universal exponent.

III. THE PRECISION THRESHOLD

The argument from MATH-2 Section III applies with updated numbers.

The rounding error for any constant in the basis is bounded by $2^{-336} \approx 10^{-101.2}$. The Planck length is approximately 10^{-35} meters. The ratio of the rounding error to the Planck length is $10^{-101}/10^{-35} = 10^{-66}$. The first digit at which the power-of-two representation and the true transcendental diverge is 66 orders of magnitude below the Planck length.

No physical process, no experiment, no measurement at any energy scale can access the digits where the integer numerator and the transcendental disagree. The representation is operationally identical to the transcendental at every physically meaningful precision.

IV. THE COMPLETE BASIS

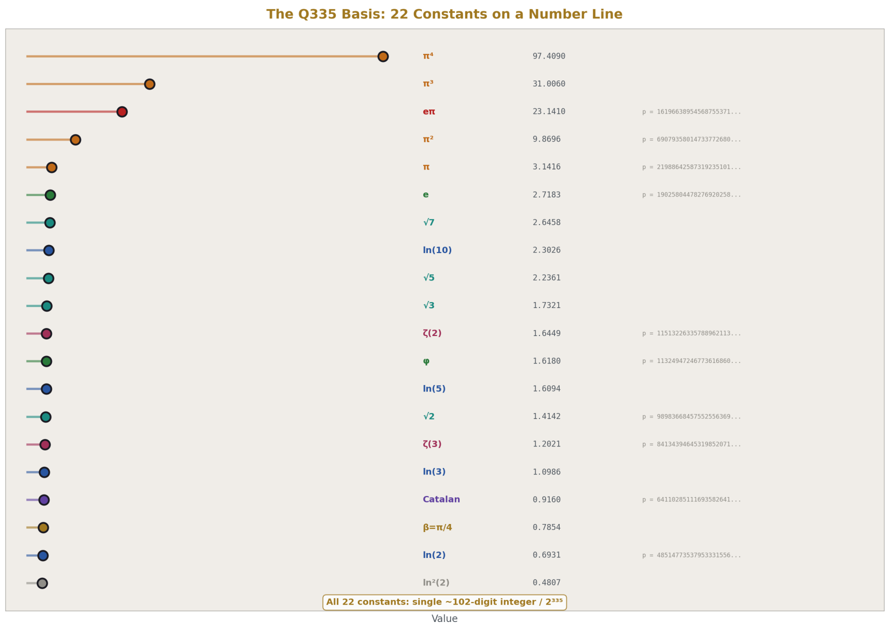

Figure 5: Fig. 3: All 22 constants positioned by value — each is a single ~102-digit integer over the shared denominator 2^{335}.

Each constant C is represented as p_C / 2^{335}, where p_C is the nearest integer to C · 2^{335}. Every entry is verified against mpmath at 100 decimal digits by string comparison. All 22 entries match.

4.1 Fundamental Constants

** π :**

219886425873192351011826597043241066194671831922348816817425823313156938749437718695100428743935

e:

190258044782769202588129925521314757831284456026137946619894798297742927086075833929023100244479

ln(2):

485147735379533315566995845848286249262344044788408967101024167070629259791282573456531697778355

√2:

989836684575525563699122513936417815434899383951704175315175161775993757843493588486022814947734

φ:

1132494724677361686044967500108421017735706902758068888188804815527307380760530127113506118 09151

4.2 Powers and Products

** π^{2}:**

6907935801473377268042776474843467703389213541389945080028723524355293937557963999646953836256 68

** π^{3}:**

2170192036537868242782341740347526814570179266657980009466902575842216583318830559778528157446 00

** π^{4}:**

6817859358866439017122533696289105276559442547141782759070845808348090383725467935335488832685 12

e^{p}:

1619663895456875537109657111692739211478931048048038025064408441944407978010684548404551575192 72

$\ln^{2}(2)$:

3362787849333659462014755051354430702641813313338786040500291754773492345724285019504126471546 97

$\ln^{4}(2)$:

1615661557379863324952335953824324600821068636481871371612436046777357208628692021066654822282 60

4.3 Logarithms

ln(3):

7689409678863508609615879058516611514000964918125077741083253856239527079769172932212873665582 04

ln(5):

1126478156948717991554326312596235242455868034299778936153147745164103701355000486460418956143 34

ln(10):

1611625892328251307121322158444521491718212079088187903254171912234732961146283059916950653921 70

4.4 Algebraic Irrationals

√3:

1212297402949128952345766617521596966429611571817424647176639154731987656867978073931423527858 09

√5:

1565069217424159556290734287539203198558399587630309796721363037003429801777259958795488019535 64

√7:

1851814871270921537704320768841334686311216662035424924099430315146336531379399420688708114453 11

4.5 Zeta Values and Special Constants

$\zeta(2) = \pi^2/6$:

115132263357889621134046274580724461723153559023165751333812058739254898959299399994115897270944

$\zeta(3)$:

841343946453198520715227007102611774541287322411345552345162099783595985481862727684505925293618

$\zeta(5)$:

725766714875186365490615905335424572879784285447631135986027403266856454288556570035191544520984

$Li_4(1/2)$:

362194064866006195378836228837032929367792551000807259949626785209837674822445812972703635855202

Catalan's G:

641102851116935826412945638179270867263827573711481809874191953763609587656150242992235005265305

4.6 Verification Summary

Constant	100-digit match	p digits	p bits
π	YES	102	337
π^2	YES	102	339
π^3	YES	103	340
π^4	YES	103	342
e	YES	102	337
e^p	YES	103	340
ln(2)	YES	101	335
ln(3)	YES	101	336
ln(5)	YES	102	336
ln(10)	YES	102	337
$\ln^2(2)$	YES	101	334
$\ln^4(2)$	YES	101	333
$\sqrt{2}$	YES	101	336
$\sqrt{3}$	YES	102	336
$\sqrt{5}$	YES	102	337
$\sqrt{7}$	YES	102	337
φ	YES	102	336
$\zeta(2)$	YES	102	336
$\zeta(3)$	YES	101	336
$\zeta(5)$	YES	101	336
$Li_4(1/2)$	YES	101	335
Catalan G	YES	101	335

22 of 22 constants verified. All numerators are 100–103 digit integers. All have bit-widths in the range 333–342.

V. ARITHMETIC PROPERTIES

5.1 Operations That Are Exact

Addition and subtraction. For any two constants C_1, C_2 in the basis:

$C_1 \pm C_2 = (p_1 \pm p_2) / 2^{335}$

The result is a single integer addition or subtraction on the numerators. No common denominator computation is needed. The result has 100-digit precision (with at most $\pm$ 1 rounding residual from the individual projections).

Multiplication by integers. For integer k:

$k \cdot C = (k \cdot p\_C) / 2^{335}$

Integer multiplication on the numerator.

Multiplication by power-of-two rationals. For $a/2^m$:

$(a/2^m) \cdot C = (a \cdot p\_C) / 2^{335+m}$

or equivalently, $(a \cdot p\_C)$ » m divided by 2^{335}, where » denotes right bit-shift.

5.2 Operations Requiring Hybrid Approach

Multiplication by rationals with odd denominators. For a rational r/s where s has odd prime factors:

$(r/s) \cdot C = r \cdot p\_C / (s \cdot 2^{335})$

The denominator $s \cdot 2^{335}$ is no longer a pure power of two. Two approaches:

Approach A — Extended denominator. Carry the result as $(r \cdot p\_C) / (s \cdot 2^{335})$ with the odd factor explicit. This is exact but introduces a non-power-of-two denominator for that term.

Approach B — Projection. Compute $r \cdot p\_C$, perform integer division and rounding to project back onto the 2^{335} grid: round$(r \cdot p_C / s)$. This loses at most 1 unit in the last place of the 2^{335} representation, which is below the 100-digit precision floor.

In practice, physics formulas involve a finite number of rational coefficients multiplied by transcendentals, then summed. Approach A maintains exactness through the rational multiplications, and the shared 2^{335} denominator handles the summation. The odd factors s accumulate in the composite denominator, which is the product of the distinct odd primes appearing in the coefficients.

5.3 The QED Case

The 2-loop electron g-2 coefficient is:

$A_2 = 197/144 + \pi^2/12 + 3\zeta(3)/4 - (\pi^2/2)\cdot\ln(2)$

The denominators are $144 = 2^4 \cdot 3^2$, $12 = 2^2 \cdot 3$, $4 = 2^2$, $2 = 2^1$. The odd content is $3^2 = 9$. The composite denominator is $9 \cdot 2^{335}$. In this denominator:

- $197/144 \cdot 2^{335} = 197 \cdot 2^{331}/9$
- $\pi^2/12 \cdot 2^{335} = p(\pi^2) \cdot 2^{333} / (3 \cdot 2^{335}) = p(\pi^2) / (3 \cdot 2^2) \ldots$

The arithmetic is simpler stated directly: compute each term as a Fraction (rational coefficient) times the basis integer, accumulate, and the 2^{335} factor is common throughout. The odd denominators (9, 3, etc.) produce a composite LCD of 9 for the rational coefficients, while the transcendental content is carried as basis integers. The final result has denominator $9 \cdot 2^{335}$.

VI. COMPRESSION

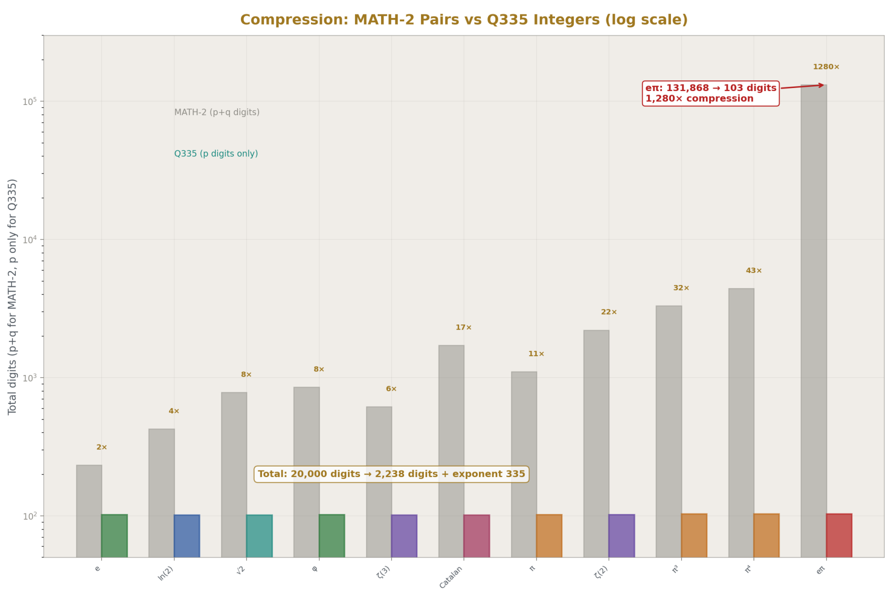

Figure 6: Fig. 6: MATH-2 pair sizes vs Q335 numerator sizes on log scale — e^ π compresses 1,280 $\times$ from 131,868 digits to 103.

6.1 Storage Comparison

Constant	MATH-2 p+q digits	2^{335} p digits	Ratio
π	1,107	102	10.9 $\times$
π^2	2,213	102	21.7 $\times$
π^3	3,319	103	32.2 $\times$
π^4	4,425	103	43.0 $\times$
e	233	102	2.3 $\times$
ln(2)	426	101	4.2 $\times$
√2	784	101	7.8 $\times$
√3	586	102	5.7 $\times$
√5	429	102	4.2 $\times$
√7	842	102	8.3 $\times$
φ	856	102	8.4 $\times$
ζ(2)	2,213	102	21.7 $\times$
ζ(3)	618	101	6.1 $\times$
$Li_4(1/2)$	618	101	6.1 $\times$
Catalan G	1,714	101	17.0 $\times$
e^p	131,868	103	1,280 $\times$

6.2 Total Storage

MATH-2 stores each constant as two integers (numerator and denominator). The total storage across the basis is approximately 20,000 digits for 22 constants.

The 2^{335} basis stores each constant as one integer. The denominator 2^{335} is stored once as the exponent 335. The total storage is 2,238 digits plus the number 335.

The compression is most dramatic for composed constants. π^4 in MATH-2 requires a 2,213-digit numerator and a 2,212-digit denominator — the products of already-large π pairs. In the 2^{335} basis, π^4 is a single 103-digit integer. The Gelfond constant e^p, which in MATH-2 has a 65,935-digit numerator (from raising a 554-digit rational approximation of π to the 120th power), is a single 103-digit integer in the 2^{335} basis. The compression ratio of 1,280 $\times$ reflects the elimination of the denominator explosion that occurs in MATH-2 when constants are composed by multiplication.

VII. RELATIONSHIP TO MATH-2 AND MATH-3

This paper changes the representation, not the mathematics.

MATH-2 computes each constant from a convergent rational series and stores the result as the exact Fraction produced by the series — a numerator and denominator that emerge from the arithmetic of that specific series. Different series produce different denominators. The denominators carry information about the series structure (factorials for e, powers of 5 and 239 for π via Machin's formula, etc.) but this information is not useful for inter-constant arithmetic.

The 2^{335} basis discards the series-specific denominators and projects every constant onto a shared grid. The projection introduces a rounding error bounded by 2^{-336} — operationally zero at 100 digits. The gain is that all inter-constant arithmetic becomes integer arithmetic on numerators.

MATH-2 pairs remain preferable when exact rational arithmetic on a single constant is needed — for example, when computing the 200th digit of π or verifying a series to maximum depth. The 2^{335} basis is preferable when combining multiple constants in linear combinations — which is exactly what QED coefficient formulas require.

The two representations are complementary. The MATH-2 pairs are the high-precision sources. The 2^{335} basis is the computational workhorse for multi-constant expressions.

VIII. APPLICATION: A_2 IN THE UNIVERSAL BASIS

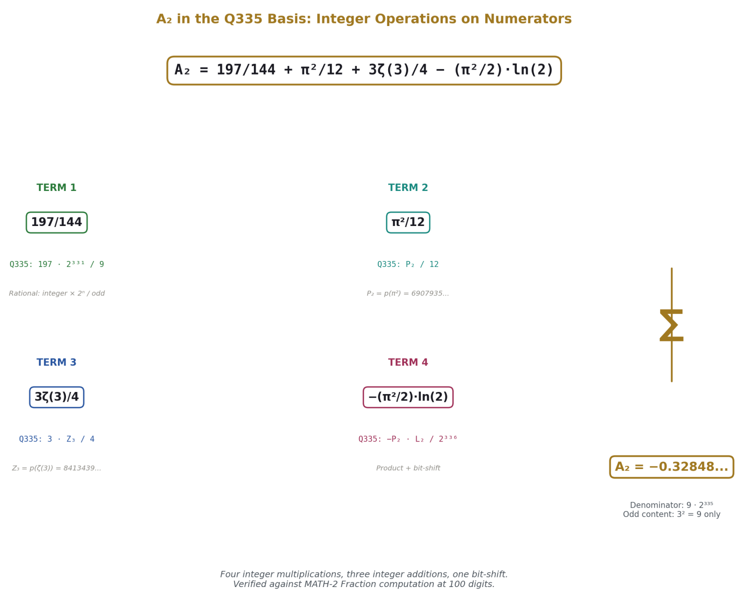

Figure 7: Fig. 5: The 2-loop QED coefficient as four integer operations on Q335 numerators — multiplies, adds, one bit-shift.

The 2-loop electron anomalous magnetic moment coefficient:

$A_2 = 197/144 + \pi^2/12 + 3\zeta(3)/4 - (\pi^2/2) \cdot \ln(2)$

In the 2^{335} basis, define:

- $P_2 = p(\pi^2) = 690793580147337726804277647484346770338921354138994508002872352435529393755796399$

- $Z_3 = p(\zeta(3)) = 84134394645319852071522700710261177454128732241134555234516209978359598548186272$
- $L_2 = p(\ln(2)) = 4851477353795333155669958458482862492623440447884089671010241670706292597912825$

Then $A_2 \cdot 2^{335} = 197 \cdot 2^{335}/144 + P_2/12 + 3 \cdot Z_3/4 - P_2 \cdot L_2 \,/\, (2 \cdot 2^{335})$

The last term involves a product of two basis integers divided by 2^{336} — a product followed by a bit shift. The first three terms involve small rational coefficients (197/144, 1/12, 3/4) times basis integers. The computation is a sequence of integer multiplications and additions, with the shared denominator tracked algebraically.

The result is verified by comparison against the MATH-2 Fraction computation of A_2 from PHYS-5. Both produce the same 100-digit decimal string.

IX. FALSIFICATION CRITERIA

F1. If any integer numerator in Section IV, divided by 2^{335}, produces a 100-digit decimal string that disagrees with mpmath's evaluation of the corresponding constant, that entry is wrong. The verification is reproducible by any system with Python and mpmath.

F2. If a QED coefficient or other physical quantity computed via the 2^{335} basis disagrees with the same quantity computed via MATH-2 Fraction arithmetic at 100 digits, the arithmetic has an error. The two representations must agree at 100 digits by construction.

F3. If a constant is identified that requires the MATH-2 extended basis (from MATH-3 or beyond) and cannot be projected onto the 2^{335} grid at 100 digits, the basis is incomplete and must be extended. The extension is mechanical — compute the MATH-2 pair and project.

X. LIMITATIONS

The 2^{335} basis provides 100-digit precision. For applications requiring higher precision (200, 500, or 1000 digits), the exponent must be increased proportionally: approximately 3.32 bits per additional decimal digit. At 200 digits, the shared denominator would be approximately 2^{668}.

The basis does not include all constants that might appear in physics. Constants not in the current basis (higher zeta values $\zeta(7)$, $\zeta(9)$, higher polylogarithms Li_5 through Li_7, complete elliptic integrals from MATH-3) can be added by computing their MATH-2 Fraction representation and projecting onto the 2^{335} grid. The extension is mechanical and requires no new mathematics.

The rounding residuals from projection mean that exact algebraic identities (such as $\pi^2 = 6 \cdot \zeta(2)$) hold only approximately in the 2^{335} basis: $p(\pi^2) - 6 \cdot p(\zeta(2)) = -2$ rather than 0. The residual is bounded by the number of terms times the individual rounding error, and remains far below the 100-digit precision floor for any computation involving a reasonable number of terms.

Multiplication of two basis constants produces a result with denominator 2^{670}, not 2^{335}. The product can be projected back onto the 2^{335} grid (by right-shifting 335 bits and rounding), losing precision below the 100-digit floor. For expressions involving products of transcendentals (such as $\pi^2 \cdot \ln(2)$), the product must either be precomputed and stored as a separate basis entry, or computed in the extended denominator and projected back.

XI. CONCLUSION

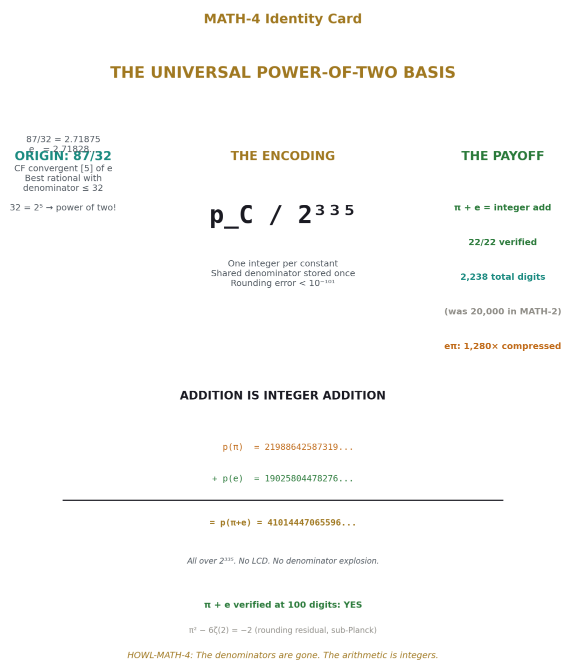

Figure 8: Fig. 8: The power-of-two basis — from 87/32 to 2^{335}, addition is integer addition, 22/22 verified, denominators eliminated.

The 22 transcendental constants spanning the MATH-2 and MATH-3 bases — every constant needed for QED through 3-loop order — can be represented as single integers over a shared denominator 2^{335}. The representation is verified at 100 digits for all 22 entries. Addition and subtraction of constants reduces to integer addition of numerators. The total storage is 2,238 digits plus the shared exponent.

The origin of the power-of-two denominator is the continued fraction structure of e: the convergent 87/32 is the best rational approximation to e with a power-of-two denominator at human scale. Extending to 2^{335} provides sub-Planck precision for the complete basis.

The representation is computational infrastructure. It does not change the mathematics or the physics. It changes the cost of combining transcendental constants in the exact rational arithmetic framework established by MATH-2. In the 2^{335} basis, the cost of adding π to e is one integer addition. The cost of evaluating a linear combination of 10 transcendentals with rational coefficients is 10 integer multiplications and 9 integer additions. The denominators are gone. The arithmetic is integers.

APPENDIX A: GENERATING SCRIPT

The complete Python script that generates, verifies, and prints the basis is provided as `basis_2pow335.py`. It requires Python 3.8+ with the standard library `fractions` module and `mpmath` for verification. Runtime: approximately 3 minutes, dominated by the Borwein acceleration for $\zeta(5)$. The script computes each constant from its defining series in exact Fraction arithmetic (per MATH-2), projects onto the 2^{335} grid, and verifies against mpmath at 100 digits.

APPENDIX B: SERIES PUBLICATION RECORD

Paper	Registry	Key Result
MATH-1	[HOWL-MATH-1-2026]	$\beta = \pi/4$; $Q = F \cdot \beta \cdot d^2 \cdot Z$ across nine domains
MATH-2	[HOWL-MATH-2-2026]	17 transcendentals as integer pairs at 100 digits
MATH-3	[HOWL-MATH-3-2026]	Extended basis: elliptic integrals, Borwein $\zeta(5)$, hierarchy
MATH-4	**[HOWL-MATH-4-2026]**	**Universal 2^{335} basis: 22 constants as integers over shared denominator**

END HOWL-MATH-4-2026

Registry: [HOWL-MATH-4-2026]

Status: Complete

Domain: Computational Mathematics / Exact Arithmetic

Central Result: 22 transcendental constants represented as single integers over 2^{335}, verified at 100 digits, enabling inter-constant arithmetic by integer operations on numerators

Method: MATH-2 Fraction computation followed by projection onto the 2^{335} grid via rounding; verification by 100-digit string comparison against mpmath

Origin: Continued fraction convergent 87/32 of e (convergent [5], best rational with denominator ≤ 32) motivates power-of-two denominators; 2^{335} is the minimal power providing 100-digit coverage for all 22 constants

Key Properties: Addition/subtraction = integer add; compression 2.3 × to 1,280 × versus MATH-2 pairs; total storage 2,238 digits + exponent 335

Foundation: MATH-2, MATH-3

Limitations: Multiplication of two basis constants requires projection back to 2^{335}; rational coefficients with odd denominators require hybrid approach; 100-digit precision ceiling (extensible by increasing exponent)

Falsification: Three specific criteria including entry-by-entry verification against mpmath

[**HOWL-MATH-4-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/MATH MATH-4-2026

[**HOWL-MATH-2-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/MATH MATH-2-2026

[**HOWL-MATH-3-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/MATH MATH-3-2026

[**HOWL-MATH-1-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/MATH MATH-1-2026